

UDP vs TCP

- UDP Header: Fixed size of 8 bytes.
- UDP Total Size Range: Minimum = 8 bytes (0 payload), Maximum = 65,535 bytes (216-1).
- UDP Max Payload Size: 65,535-8=65,527 bytes.
- TCP Header Range: Minimum = 20 bytes (HLEN = 5), Maximum = 60 bytes (HLEN = 15).
- HLEN Multiplier: Multiply HLEN by 4 to get bytes (HLEN×4=Bytes).
- In a 16-hex-character UDP dump (0632 000D 001C E217):
- 0632 (Chars 1-4) = Source Port
- 1586
- 000D (Chars 5-8) = Destination Port
- 001C (Chars 9-12) = Total Length
- E217 (Chars 13-16) = Checksum

Data length = total length - header size

Source port, destination port. LOW : server, HIGH : user

Here are the direct and comprehensive answers to all the questions from your uploaded practice set:

1) What is the minimum and maximum size of a UDP datagram?

- Minimum Size: 8\$ bytes
 - Explanation: A UDP header is fixed at 8\$ bytes long. When a datagram carries no payload (0 bytes of process data), the minimum size is simply the size of the header itself.
- Maximum Size: 65,535\$ bytes
 - Explanation: The UDP header contains a 16-bit Length field. The largest number that can be expressed with 16 bits in binary is $2^{16} - 1 = 65,535$.

2) What is the minimum and maximum size of the process data that can be encapsulated in a UDP datagram?

- Minimum Process Data Size: 0\$ bytes
 - Explanation: Applications can send an empty message (containing only the 8-byte UDP header) for signaling or heartbeat purposes.
- Maximum Process Data Size: 65,527\$ bytes
 - Explanation: Subtract the mandatory 8-byte UDP header from the total maximum datagram size:
 - $\text{Max Process Data} = 65,535 \text{ bytes (Max UDP Datagram)} - 8 \text{ bytes (UDP Header)} = 65,527 \text{ bytes}$

3) What is the maximum size of the TCP header? What is the minimum size of the TCP Header?

- Minimum Size: 20\$ bytes
 - Explanation: When no optional fields are attached, the basic TCP header is exactly 20\$ bytes long.
- Maximum Size: 60\$ bytes
 - Explanation: The TCP header includes a 4-bit field called Header Length (HLEN) or Data Offset. The highest value a 4-bit number can store is 15\$ in decimal (1111_2). Since HLEN counts in 4-byte words:
 - $\text{Max Header Size} = 15 \times 4 \text{ bytes} = 60 \text{ bytes}$

4) UDP Header Hex Dump Analysis

Hex Dump: 0632 000D 001C E217

A UDP header consists of four 2-byte (4 hex characters) fields:

1. 0632 → Source Port
 2. 000D → Destination Port
 3. 001C → Total Length
 4. E217 → Checksum
- a. What is the source port number?
 - Answer: 1586\$
 - Calculation: Convert 0632 from Hex to Decimal:
 - $0 \times 16^3 + 6 \times 16^2 + 3 \times 16^1 + 2 \times 16^0 = 1536 + 48 + 2 = 1586$
 - b. What is the destination port number?
 - Answer: 13\$
 - Calculation: Convert 000D from Hex to Decimal: D in hex equals 13\$.
 - c. What is the total length of the user datagram?
 - Answer: 28\$ bytes
 - Calculation: Convert 001C from Hex to Decimal:
 - $1 \times 16^1 + 12 \times 16^0 = 16 + 12 = 28$
 - d. What is the length of the data?
 - Answer: 20\$ bytes
 - Calculation: $\text{Total Length} - \text{Header Size} = 28 \text{ bytes} - 8 \text{ bytes} = 20 \text{ bytes}$
 - e. Is the packet directed from a client to a server or vice versa?
 - Answer: Client to Server
 - Explanation: Destination Port 13\$ is a well-known port reserved for servers (specifically the Daytime protocol). Source Port 1586\$ is an ephemeral (dynamic) port assigned temporarily by a client process.
 - f. What is the client process?
 - Answer: Daytime protocol
 - Explanation: Port 13\$ is assigned by IANA for the Daytime service, which sends back the current date and time to the requesting client.
- 5) Why do we need a DNS system when we can directly use an IP address?
- Human Usability: Humans easily remember textual domain names (like google.com), whereas remembering numerical IPv4 addresses (142.250.190.46) or complex IPv6 addresses is difficult.
 - IP Abstraction & Flexibility: Web servers and hosting locations frequently change their physical IP addresses (e.g., due to migrations or load balancing). DNS acts as a mapping layer so that when an IP address changes behind the scenes, users can still access the service using the exact same domain name without interruption.
- 6) In Symmetric-key cryptography, can Alice use the same key to communicate with both Bob and John? Explain your answer.
- Answer: No, she should not.
 - Explanation: In symmetric-key cryptography, the exact same secret key is used to both encrypt and decrypt messages. If Alice uses the same secret key \$K\$ for both Bob and John, then Bob possesses key \$K\$. With this key, Bob can decrypt all private communications sent between Alice and John, and Bob can also impersonate Alice when sending messages to John. To maintain confidentiality, Alice must share a unique secret key with each person.
- 7) In Asymmetric-key cryptography, can Alice use the same key to communicate with both Bob and John? Explain your answer.
- Answer: Yes.
 - Explanation: Asymmetric cryptography uses a mathematically linked key pair: a Public Key (which can be distributed to everyone) and a Private Key (kept secret by the owner).
 - If Bob and John want to send encrypted messages to Alice, both Bob and John encrypt their messages using Alice's Public Key. Only Alice can decrypt these messages using her single matching Private Key.
 - If Alice wants to send encrypted messages to Bob and John, she uses Bob's Public Key to encrypt data meant for Bob, and John's Public Key for data meant for John.
- 8) A hash algorithm creates a digest of \$N\$ bits. How many different digests can be created from this algorithm?
- Answer: 2^N different digests
 - Explanation: Each bit in binary can hold 2 possible values (\$0\$ or \$1\$). For an output digest length of \$N\$ bits, the total number of unique binary combinations (different possible output digests) is $2 \times 2 \times \dots (N \text{ times}) = 2^N$. $\text{Digests} = 2^N$
- 9) Explain the difference between a stateless firewall and a stateful firewall.
- Stateless Firewall: Inspects each incoming or outgoing packet in complete isolation based strictly on static rules (such as source/destination IP address and port number) without any memory of previous traffic. It requires explicit rules configured for both inbound and outbound directions.
 - Stateful Firewall: Keeps track of active network connections over time using a State Table. It categorizes packets based on connection context (NEW, ESTABLISHED, RELATED) and automatically permits returning response traffic for legitimate, established connections initiated from inside the network.